

Module 01: Systems in Robotics

Learning Objectives

1. Define **Systems** within the context of Robotics.
2. Analyze how **Systems** interacts with other components of the Active Inference framework.
3. Apply specific constraints of Robotics to the formal definition of **Systems**.

Introduction

This module explores **Systems**. In the **Robotics** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. **Systems** is a critical component of the 8-part Active Inference spine, bridging the gap between [Previous Topic] and [Next Topic].

Key Concepts

1. Systems as a Markov Blanket Boundary

How does **Systems** define the boundary between the agent and the environment?

2. Generative Models of Systems

What parameters involved in **Systems** must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of **Systems** drive the perception-action loop?

Applications

In Robotics, we see **Systems** manifest in: * **Specific Example 1:** [Add domain-specific example here] * **Specific Example 2:** [Add domain-specific example here]

Conclusion

Understanding Systems allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.